

Arjuna JEE 2027

Maths by Amarnath Anand Sir

Quadratic Equations

DPP: 9

Total Duration - 35 Min

- Q1** If $a < b < c < d$, and $k > 0$ then the quadratic equation $(x - a)(x - c) + k(x - d) = 0$ has
(A) All roots real and distinct
(B) All roots real but not necessarily distinct
(C) All roots real and negative
(D) May be imaginary
- Q2** If $a + b + c > \frac{9c}{4}$ and the equation $ax^2 + 2bx - 5c = 0$ has non-real complex roots, then
(A) $a > 0, c > 0$
(B) $a > 0, c < 0$
(C) $a < 0, c < 0$
(D) $a < 0, c > 0$
- Q3** If exactly one root of the equation $x^2 - 2kx + k^2 - 1 = 0$ satisfies the inequality $\log_{\sqrt{3}}(2 - x) \leq 0$ then find sum of all possible integral values of k .
- Q4** If α, β & γ are the roots of the equation $x^3 - x - 1 = 0$ then, $\frac{1+\alpha}{1-\alpha} + \frac{1+\beta}{1-\beta} + \frac{1+\gamma}{1-\gamma}$ has the value equal to:
(A) zero (B) -1
(C) -7 (D) 1
- Q5** If α, β are the roots of the equation $ax^2 + bx + c = 0$, then the roots of the equation $a(2x + 1)^2 + b(2x + 1)(x - 1) + c(x - 1)^2 = 0$ are
(A) $\frac{2\alpha+1}{\alpha-1}, \frac{2\beta+1}{\beta-1}$
(B) $\frac{2\alpha-1}{\alpha+1}, \frac{2\beta-1}{\beta+1}$
(C) $\frac{\alpha+1}{\alpha-2}, \frac{\beta+1}{\beta-2}$
(D) $\frac{2\alpha+3}{\alpha-1}, \frac{2\beta+3}{\beta-1}$
- Q6** Let $f(x) = x^3 + x + 1$ and $P(x)$ be a cubic polynomial such that $P(0) = -1$ and the roots of $P(x) = 0$ are the squares of the roots of $f(x) = 0$, then the value of $P(9)$ is-
(A) 98 (B) 899
(C) 80 (D) 898
- Q7** If α & β ($\alpha < \beta$) are the roots of the equation $x^2 + bx + c = 0$ where $c < 0 < b$, then
(A) $0 < \alpha < \beta$
(B) $\alpha < 0 < \beta^2 < \alpha^2$
(C) $\alpha < \beta < 0$
(D) $\alpha < 0 < \alpha^2 < \beta^2$
- Q8** If ' x ' is real and $k = \frac{x^2-x+1}{x^2+x+1}$, then:
(A) $\frac{1}{3} \leq k \leq 3$
(B) $k \geq 5$
(C) $k \leq 0$
(D) None of these
- Q9** If both the roots of the quadratic equation $x^2 - 2kx + k^2 + k - 5 = 0$ are less than 5, then k lies in the interval
(A) $(-\infty, 4)$
(B) $[4, 5]$
(C) $(5, 6]$
(D) $(6, \infty)$
- Q10** The values of a for which $2x^2 - 2(2a + 1)x + a(a + 1) = 0$ may have one root less than a and other root greater than a are given by
(A) $1 > a > 0$
(B) $-1 < a < 0$
(C) $a \geq 0$
(D) $a > 0$ or $a < -1$
- Q11** All the values of m for which both roots of the equation $x^2 - 2mx + m^2 - 1 = 0$ are greater than -2 but less than 4 , lie in the interval



- (A) $-2 < m < 0$
 (B) $m > 3$
 (C) $-1 < m < 3$
 (D) $1 < m < 4$

Q12 If α, β are the roots of $x^2 - 3x + a = 0, a \in \mathbb{R}$ and $\alpha < 1 < \beta$ then

- (A) $a \in (-\infty, 2)$
 (B) $a \in (-\infty, \frac{9}{4}]$
 (C) $a \in (2, \frac{9}{4}]$
 (D) none of these

Q13 If α, β are roots of equation $3x^2 - 9x - \lambda = 0, \lambda \in \mathbb{R}$ such that $1 < \alpha < 3$ & $3 < \beta < 5$ then λ lies in

- (A) $(-6, 30)$ (B) $(-6, 0)$
 (C) $(0, 30)$ (D) Null set

Q14 Suppose a, b denotes the distinct real roots of the quadratic polynomial $x^2 + 20x - 2020$ and suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2020$.

Then the value of ac

$$(a - c) + ad(a - d) + bc(b - c) + bd$$

$(b - d)$ is

- (A) 0 (B) 8000
 (C) 8080 (D) 16000

Q15 If α, β are the roots of $x^2 - 3x + 1 = 0$ then the equation whose roots are $\left(\frac{1}{\alpha-2}, \frac{1}{\beta-2}\right)$ is

- (A) $x^2 + x - 1 = 0$
 (B) $x^2 + x + 1 = 0$
 (C) $x^2 - x - 1 = 0$
 (D) None of these

Q16 The quadratic equation whose roots are reciprocal of the equation $ax^2 + bx + c = 0$ is

- (A) $cx^2 + bx + a = 0$
 (B) $bx^2 + cx + a = 0$
 (C) $cx^2 + ax + b = 0$
 (D) $bx^2 + ax + c = 0$



Answer Key

Q1 (A)

Q2 (B)

Q3 $k = 0, 2$

Q4 (C)

Q5 (C)

Q6 (B)

Q7 (B)

Q8 (A)

Q9 (A)

Q10 (D)

Q11 (C)

Q12 (A)

Q13 (D)

Q14 (D)

Q15 (C)

Q16 (A)



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